Pipeline Test Sheet

Pipeline test sheet: every charter component, numbers read from a CSV file

MindShelf

1 Core formulas

Desired over have

$$V = \frac{D}{H} \times Q$$

D ordered dose, H dose on hand, Q volume containing H.

Gravity drip rate

$$\text{rate} = \frac{V \times \text{DF}}{T \times M}$$

V volume (mL), DF drop factor (gtt/mL), T duration (h), M minutes per hour.

Hourly rate

$$\text{rate (mL/h)} = \frac{V\ \text{(mL)}}{T\ \text{(h)}}$$

1.1 Worked examples

EXAMPLE Oral suspension

Ordered 250 mg; on hand 500 mg per 5 mL.

$$V = \frac{250}{500} \times 5 = 2.5\ \text{mL}$$

EXAMPLE Gravity infusion

1,000 mL over 8 h, drop factor 15 gtt/mL.

$$\frac{1{,}000 \times 15}{8 \times 60} = 31.25 \Rightarrow 31\ \text{gtt/min}$$

EXAMPLE Weight-based dose

Ordered 15 mg/kg; weight 12 kg.

$$15 \times 12 = 180\ \text{mg}$$

TRAP

Convert **before** you divide: mg and µg differ by a factor of 1,000.

TRAP Rounding

Round only the final result, never the intermediate steps.

2 Reference values

Case	Formula	Result
EX01	`D / H * Q`	2.5 mL
EX02	`V * DF / (T * M)`	31.25 ⇒ 31 gtt/min
EX03	`V / T`	125 mL/h
EX04	`P * W`	180 mg
EX05	`C * K`	500 µg
EX06	`C / K`	12.5 mg

2.1 Number formatting

- Thousands: 12,500 µg = 12.5 mg
- Decimals: 0.5 mg = 500 µg
  - Second-level item, rounded with `decimales: 1`: 125.0 mL/h

CLINICAL NOTE

Independent double check for high-alert medications (insulin, heparin, concentrated potassium chloride IV).

Method in 3 steps

1. Write the formula with units.
2. Convert to the same unit.
3. Check the order of magnitude.

2.1.1 Level-3 heading

Body text in Fira Sans. A external link is underlined; **bold** is semi-bold.

3 Three-column block

EXAMPLE Pump rate

$$\frac{500\ \text{mL}}{4\ \text{h}} = 125\ \text{mL/h}$$

EXAMPLE mg to µg

$$0.5\ \text{mg} \times 1{,}000 = 500\ \mu\text{g}$$

EXAMPLE µg to mg

$$\frac{12{,}500\ \mu\text{g}}{1{,}000} = 12.5\ \text{mg}$$

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